

作业 1 参考答案及评分说明

Junda Wu, Student ID: 210320621 → 25S065010

April 15, 2026

满分 30 分。A+ 是 27-30; A 是 24-27 (不含 27); A- 是 21-24 (不含 24), 然后是 B+, B……总之 3 分一档。

1. 【共 7.5 分】 Justify each answer. Draw $\checkmark$ if True and $\times$ if False. 每题 0.5 分, 不需要写理由。

题号	a	b	c	d	e	f	g	h	i	j1	j2	j3	j4	j5	j6
答案	$\checkmark$	$\times$	$\times$	$\checkmark$	$\times$	$\times$	$\times$	$\checkmark$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\checkmark$	$\times$

2. 【共 5 分】 Let W be the union of the first and third quadrants in the xy -plane. That is, let $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : xy \geq 0, x, y \in \mathbb{R} \right\}$.

(a) If $\mathbf{u}$ is in W and c is any scalar, is $c\mathbf{u}$ in W ? Why?

(b) If $\mathbf{u}, \mathbf{v} \in W$, is $\mathbf{u} + \mathbf{v}$ in W ? Why?

(c) Is W a vector space?

Solution:

(a) Yes. $c\mathbf{u} = [cx \quad cy]^T$. Since $\mathbf{u} = [x \quad y]^T \in W, xy \geq 0$. Thus $(cx)(cy) = c^2xy \geq 0$. Therefore $c\mathbf{u} \in W$. (2 分, 判断、证明各 1 分。判断不对则全错)

(b) No. Counterexample: $\mathbf{u} = [2 \quad 1]^T \in W, \mathbf{v} = [-1 \quad -2]^T \in W$, but $\mathbf{u} + \mathbf{v} = [1 \quad -1]^T \notin W$. (2 分, 判断、证明各 1 分。判断不对则全错)

(c) No, because the addition in W is not closed. (1 分, 只要有判断即可)

3. 【共 6 分】 Let X and Y be subspaces of a vector space V .

(a) Let $\mathbf{v} \in V, \mathbf{v} \notin X$. Prove that if $\mathbf{x} \in X$ then $\mathbf{x} + \mathbf{v} \notin X$.

(b) Use (a) to show that $X \cup Y$ is a subspace if and only if $X \subset Y$ or $Y \subset X$.

(c) Give an example that X and Y are both subspaces of a vector space V , but $X \cup Y$ is not.

Solution:

(a) Suppose that $\mathbf{v} \in V, \mathbf{v} \notin X$, and $\mathbf{x} \in X$. If $\mathbf{x} + \mathbf{v} \in X$, then since X is a subspace, $\mathbf{v} = (\mathbf{x} + \mathbf{v}) - \mathbf{x} \in X$, which is a contradiction. Therefore, $\mathbf{x} + \mathbf{v} \notin X$. (2 分, 看错误程度扣分) $\square$

(b) If $X \subset Y$ or $Y \subset X$, then $X \cup Y = X$ ($Y \subset X$) or $X \cup Y = Y$ ($X \subset Y$). Since X and Y are subspaces, $X \cup Y$ is also a subspace.

For the converse, it will suffice to show that if $X \not\subset Y$ and $Y \not\subset X$, then $X \cup Y$ is not a subspace. Since $X \not\subset Y$ and $Y \not\subset X$, there exist $\mathbf{x}, \mathbf{y} \in X \cup Y$ such that $\mathbf{y} \in Y, \mathbf{y} \notin X$ and $\mathbf{x} \in X, \mathbf{x} \notin Y$. Substitute $\mathbf{x}$ and $\mathbf{y}$ into $\mathbf{x}$ and $\mathbf{v}$ in (a), we know that $\mathbf{x} + \mathbf{y} \notin X$; interchange the role of $\mathbf{x}$ and $\mathbf{y}$ we get that $\mathbf{x} + \mathbf{y} \notin Y$. Hence $X \cup Y$ is not closed under addition, and it is not a subspace. (2 分, 两个方向各 1 分) $\square$

(c) Consider two subspaces of $\mathbb{R}^3$:

$$X = \{(x, 0, 0), x \in \mathbb{R}\}, Y = \{(0, y, 0), y \in \mathbb{R}\}.$$

X includes all vectors parallel to x -axis while Y includes all vectors parallel to y -axis. Their union includes all vectors parallel to either x -axis or y -axis. However, consider the following two vectors in $X \cup Y$:

$$\mathbf{x} = (1, 0, 0), \quad \mathbf{y} = (0, 1, 0),$$

add them together and it turns out $\mathbf{x} + \mathbf{y} = (1, 1, 0)$, which is parallel to neither the x -axis nor the y -axis, and thus is not in $X \cup Y$. (2 分, 举例需要满足: X 和 Y 确实是子空间; $X \cup Y$ 确实不是子空间。这两点各 1 分。比如, 如果举上一题的例子, 就只有 1 分, 因为上一题的用于求并集的集合不是子空间)

4. 【共 2 分】 Prove or give a counterexample: If X, Y, W are subspaces of V such that $X \oplus W = Y \oplus W$, then $X = Y$.

Solution: Counterexample: $X = \text{span}\{(1, 0, 0), (0, 1, 0)\}$, $Y = \text{span}\{(1, 0, 1), (0, 1, 0)\}$, and $W = \text{span}\{(0, 0, 1)\}$. $X \oplus W = Y \oplus W = \mathbb{R}^3$, but $X \neq Y$. (2 分, 判断不对就全错; 判断正确看举出的反例, 举例不对 [例如不是直和、不是子空间] 扣 1 分)

5. 【共 2.5 分】 Suppose that V is a finite dimensional vector space. Let a system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r \in V$ be linearly independent but not generating. Show that it is possible to find a vector $\mathbf{v}_{r+1} \in V$ such that the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent. (Hint: Take for $\mathbf{v}_{r+1}$ any vector that cannot be represented as a linear combination $\sum_{k=1}^r \alpha_k \mathbf{v}_k$ and show that the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent.)

Proof:

Suppose $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ are linearly independent but do not generate V . Then there exists a vector $\mathbf{v}_{r+1} \in V$ that cannot be expressed as a linear combination $\sum_{k=1}^r \alpha_k \mathbf{v}_k$. Choose such a $\mathbf{v}_{r+1}$. We show that the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent.

Assume that $\alpha_1, \dots, \alpha_r, \alpha_{r+1}$ are scalars such that

$$\alpha_1 \mathbf{v}_1 + \dots + \alpha_r \mathbf{v}_r + \alpha_{r+1} \mathbf{v}_{r+1} = \mathbf{0}.$$

If $\alpha_{r+1} = 0$, then $\alpha_1 \mathbf{v}_1 + \dots + \alpha_r \mathbf{v}_r = \mathbf{0}$, and by linear independence of $\mathbf{v}_1, \dots, \mathbf{v}_r$ we obtain $\alpha_1 = \dots = \alpha_r = 0$. If $\alpha_{r+1} \neq 0$, then we can solve

$$\mathbf{v}_{r+1} = -\frac{\alpha_1}{\alpha_{r+1}} \mathbf{v}_1 - \dots - \frac{\alpha_r}{\alpha_{r+1}} \mathbf{v}_r,$$

which contradicts the choice of $\mathbf{v}_{r+1}$. Therefore all coefficients are zero, and the vectors are linearly independent. (2.5 分, 酌情给分, 只要不特别离谱, 就至少给 1 分; 但不能纵容错误) $\square$

6. 【共 7 分】 Let $U = \{p \in \mathcal{P}_4(\mathbb{F}) : p(2) = p(5)\}$, $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$.

- Give the standard basis of $\mathcal{P}_4(\mathbb{F})$ and verify that it is indeed a basis. What is the dimension of $\mathcal{P}_4(\mathbb{F})$? What is the dimension of $\mathbb{F}^5$? (Hint: Can you see the relationship between them?)
- Find a basis of U . (Hint: Try to answer: Is $1 \in U$? Is $(x-2)(x-5) \in U$? Are there any other “independent” vectors? What is the possible size of the linearly independent set?)
- Find a subspace W of $\mathcal{P}_4(\mathbb{F})$ such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$. (Hint: The proof of Theorem 2.6.3 might help.)

Solution:

(1) The standard basis of $\mathcal{P}_4(\mathbb{F})$ is

$$\{1, x, x^2, x^3, x^4\}.$$

Linear independence: If $a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 = 0$ (the zero polynomial), then all coefficients $a_i = 0$ because two polynomials are equal if and only if their coefficients coincide.

Spanning: Every $p(x) \in \mathcal{P}_4(\mathbb{F})$ can be uniquely written as $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$, so the set spans $\mathcal{P}_4(\mathbb{F})$.

Hence $\dim \mathcal{P}_4(\mathbb{F}) = 5$. The dimension of $\mathbb{F}^5$ is also 5.

Stack the coefficients of $p \in \mathcal{P}_4(\mathbb{F})$ into a column and we get a vector in $\mathbb{F}^5$; regard the scalars of a vector in $\mathbb{F}^5$ as the coefficients a polynomial and we get a $p \in \mathcal{P}_4(\mathbb{F})$. We say that the two spaces are **isomorphic** (同构) via the coefficient map $(a_0, \dots, a_4) \leftrightarrow a_0 + a_1x + \dots + a_4x^4$. (You will know this concept in the next chapter.)

(共 3 分: 写出基 1 分; 验证两个方面各 0.5 分; 写出两个维数各 0.5 分。对于思考题, 有回答者可酌情加 0.5-1 分)

(2) We look for linearly independent polynomials satisfying $p(2) = p(5)$:

- The constant polynomial 1 satisfies $1(2) = 1(5) = 1$, so $1 \in U$.
- $q(x) = (x-2)(x-5)$ satisfies $q(2) = q(5) = 0$, hence $q \in U$.
- $xq(x) = x(x-2)(x-5)$ and $x^2q(x) = x^2(x-2)(x-5)$ also vanish at 2 and 5, so they belong to U .

These four polynomials have degrees 0, 2, 3, 4 respectively; distinct degrees imply linear independence, which further implies that $\dim U \geq 4$.

Since $U \neq \mathcal{P}_4(\mathbb{F})$, $\dim U < \dim \mathcal{P}_4(\mathbb{F})$. Thus $\dim U = 4$. Therefore a basis of U :

$$\{1, (x-2)(x-5), x(x-2)(x-5), x^2(x-2)(x-5)\}.$$

(2 分。需要注意确实给出的是一个基。如果只有答案没有过程，扣 1.5 分；如果没有说明线性无关的理由，或者从“线性无关组”直接跳到基，这两项都是扣 0.5 分（可叠加）；直接构造，而不是像上面一样列举出几个代表向量，正确的也可给满分。）

(3) Since $\dim \mathcal{P}_4(\mathbb{F}) = 5$ and $\dim U = 4$, we need a one-dimensional subspace W such that $W \cap U = \{0\}$. Take for instance $W = \text{span}\{x\}$.

(Method 1): Add x to the basis of U and we get a list

$$\{1, x, (x-2)(x-5), x(x-2)(x-5), x^2(x-2)(x-5)\}.$$

These polynomials have degrees 0, 1, 2, 3, 4 respectively; distinct degrees imply linear independence. Linearly independent list of the right length (dimension) is a basis, and thus we complete the basis of U to a basis of the whole space $\mathcal{P}_4(\mathbb{F})$. From Theorem 2.6.3, we get that W is the subspace such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

(Method 2): Because $x(2) = 2 \neq 5 = x(5)$, we have $x \notin U$; if $cx \in U$ with $c \neq 0$, then $x \in U$, which is a contradiction. Hence $W \cap U = \{0\}$. Dimensions add ($4 + 1 = 5$), so $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

Thus one possible subspace is

$$W = \{ax : a \in \mathbb{F}\}.$$

(Other choices, e.g. $W = \text{span}\{x^4\}$, work as well.) (2 分。注意给出的不能落在原向量空间 U 中，否则判错，扣 1.5-2 分；如果只给出答案而没有过程，扣 1.5 分)